

# ADEKOYA JOHN A.

Senior Full-Stack Engineer · React · TypeScript · Laravel · Node.js (NestJS) · Fintech

Lagos, Nigeria (Remote-first) · +234 912 294 4976 · adebisimartins3@gmail.com · GitHub · LinkedIn · Portfolio

## PROFESSIONAL SUMMARY

Full-stack engineer with 5 years designing and shipping financial platforms end to end — payroll, invoicing, tax compliance, workforce operations, and prepaid utility payments. I own features from database schema and API design through to polished React/TypeScript interfaces. Recent highlights: founded and architected a multi-tenant e-invoicing compliance platform (cryptographic signing, three pluggable tax-authority submission drivers); led the ground-up Go rewrite of a prepaid-electricity payments platform — 14 bounded contexts, a double-entry ledger, hardware token vending — and rebuilt its four-app frontend as one Turborepo monorepo. A BSc in Accounting gives me rare domain fluency in payroll, tax computation, and reconciliation — I speak both engineer and finance.

## CORE SKILLS

**Frontend:** React · Next.js · TypeScript · TailwindCSS · shadcn/ui · Inertia.js · Vite · TanStack Query & Table · Redux Toolkit · Zustand · Storybook

**Backend:** PHP (Laravel) · Go (chi · GORM · River jobs) · Node.js (NestJS, Express) · Prisma · REST API design · OAuth2 · BullMQ/Redis queues · webhook design (HMAC verification, idempotency)

**Data:** PostgreSQL · MySQL · relational schema design · reporting & aggregation queries

**Fintech & Compliance:** statutory e-invoicing & digital signing (RSA/SHA-256) · payment gateways (collections & disbursements) · multi-currency (NGN · ZAR · USD) · withholding tax / VAT · maker-checker workflows · double-entry ledgers · STS prepaid-meter token vending · audit trails

**Architecture & DevOps:** modular monolith & clean architecture (enforced boundaries) · Feature-Sliced Design · design systems · Docker Compose · Nginx + wildcard TLS · Cloudflare · server hardening

**Quality & Observability:** Pest · Go table-driven & integration tests · property-based testing · Vitest · Jest · React Testing Library · Sentry · GitHub Actions CI/CD · Lighthouse · CSP/OWASP hardening

## PROFESSIONAL EXPERIENCE

### Full-Stack Engineer

*Stransact (RSM International — Nigeria Correspondent) · Oct 2022 – Present · Lagos, Nigeria*

Lead engineer across the firm's product portfolio: enterprise payroll (iPaySuite), tax-compliance infrastructure (NRS), workforce operations (TimeX), and a prepaid-electricity payments platform (GridCore). I own modules end to end — schema, APIs, and React frontends — and act as a repo maintainer, reviewing and merging 130+ pull requests.

### NRS Compliance — Multi-Tenant E-Invoicing Platform · Laravel 12 · React 19 · Inertia · PostgreSQL

Compliance infrastructure connecting business accounting systems to the national tax authority's mandatory e-invoicing system.

- Founded and architected the platform — sole author of the original codebase (110 commits, 326 files), clean-architecture layering with tool-enforced dependency boundaries; still own ~30% of the code after the team tripled it.
- Built the cryptographic core — RSA SHA-256 signing with deterministic payload canonicalization, spec-compliant QR encryption, invoice reference number (IRN) generation — plus a pluggable submission-driver architecture: three interchangeable drivers behind one contract, with a full response-code taxonomy, exponential-backoff retries, and forward-only webhook status transitions.
- Integrated Xero, Zoho Books, and QuickBooks (OAuth2 SSO, signature-verified webhooks, automatic token refresh), normalizing provider invoices into one canonical hash-addressed schema guarded by idempotency keys and an append-only audit ledger.
- Shipped 81 automated tests including 14 property-based suites over signing, encryption, and idempotency invariants; authored the 17-section tax-authority API integration guide the team builds from.

### iPaySuite — Enterprise Payroll, Invoicing & Payments Suite · Laravel · React · TypeScript

Used by ~20 corporate clients — processing over ₦2.1 billion (~\$1.4M) in payroll monthly, with hundreds of employees per organisation.

- Designed and built a self-contained invoicing module from schema to UI: multi-currency invoices with frozen exchange rates, credit notes, withholding tax on both receivables and payables, and tax-inclusive pricing that persists net values for regulator-accurate submissions.
- Integrated the national e-invoicing mandate — readiness validation, submission, cryptographically signed QR codes on every invoice — and built a Zoho-class template customizer with reusable presets and a single renderer for on-screen and PDF output.
- Architected a bulk disbursement module processing ₦20M+ weekly across salary, vendor, and NGO aid/beneficiary payouts: maker-checker approvals with recall, per-batch audit views, pay-period locking, and granular permission gating — full lifecycle covered by automated tests.

- Integrated a licensed payment gateway (Monnify) for both inbound invoice collections and outbound disbursements on a pooled custodian-wallet model.
- Defined the repo-wide modular-monolith architecture and the islands frontend with its shadcn/ui + Tailwind design system; built payroll approval workflows and dashboards — 10,000+ record tables at 60fps via virtualisation, Lighthouse 58 → 87.

**GridCore — Prepaid Electricity Vending & Payments Platform** · *Go* · *PostgreSQL* · *Redis* · *React* · *TypeScript* · *Turborepo* · [dev.gridcoreinc.com](https://dev.gridcoreinc.com)

Meter vending, wallets, and settlements for operators, merchants, and their customers — rewritten in 2026 from NestJS/Next.js to Go + a React monorepo (feature-complete, cutover rehearsed).

- Led the ground-up rewrite of the backend in Go — ~94k lines, 14 bounded contexts in clean-architecture rings, 58 tables: a vend-order state machine with a three-transaction pipeline, an STS hardware-security-module anticorruption layer with per-host circuit breakers and failover, Paystack collections/webhooks/payouts, a partner API (hashed bearer keys, IP allowlists, GCRA rate limits, sandbox), RBAC over 57 permissions, MFA sessions, and an audit log.
- Designed the double-entry ledger — balanced journal entries idempotent by reference, trial balance enforced as a standing test — and reconciled legacy history into an opening journal for 149 wallets, corroborated against Paystack to the kobo.
- Built the cutover tooling: nine rehearsed backfill programs, GORM + Atlas migrations, River in-database jobs inserted in the same transaction as the event, OpenAPI-generated handlers (160+ operations), 499 Go tests with in-process fakes for the HSM, Paystack and AMR, a CI gate (vet, staticcheck, drift checks) and CD to Ubuntu with runbooks.
- Rebuilt the frontend estate as one Turborepo monorepo (~35k lines of TypeScript): an RBAC-gated operator + merchant console, a white-label customer portal, and an Astro marketing site sharing a shadcn/ui design system, a typed API client mapping RFC 9457 field errors onto inputs, and lint-enforced import boundaries — 100+ Vitest/RTL tests.
- Earlier, on the legacy stack: architected the South Africa multi-currency expansion (data-driven currency config, pluggable provider registry — Paystack → NGN, Blu-Voucher → ZAR); cut meter analytics from a hardcoded 13–20 s wait to milliseconds via a BullMQ background poller; hardened the STS Thrift gateway with multi-host failover.

**TimeX — Workforce Operations & Performance Platform** · *Laravel 12* · *Inertia* · *React 18* · *TypeScript*

- #1 frontend contributor over 3 years — 583 commits and ~52% of the surviving frontend codebase (3× the next contributor) — while serving as a repo maintainer merging 130+ PRs.
- Built the leave and attendance modules end to end: a three-tier approval state machine (supervisor → HOD → HR), an entitlement engine with queued year-end roll-over, and QR-code clock-in with IP allow-listing and geofencing.
- Built the multi-entry timesheet grid with debounced cell autosave, bulk approval, and approved-row locking; led the Performa performance-review suite — dynamic form builder with autosave, period-end reviews, development plans, and weighted-average scoring.
- Drove frontend architecture: Feature-Sliced Design migration with lint-enforced layer boundaries, TanStack Query adoption, Sentry + error boundaries, a 31-story Storybook design system, and an LLM-powered timelog summary feature.

## Frontend Engineer

*One Purse Inc.* · *Feb 2022 – Oct 2022* · *Remote* — *United States*

- Led the frontend build of a Next.js e-commerce platform (catalogue, checkout, admin) supporting thousands of SKUs, and a vehicle auction marketplace with real-time bidding; lifted Core Web Vitals to green across all pages.

## Frontend Engineer (Volunteer)

*Fountain of Hope International* · *Aug 2022 – Present*

- Built React event-registration platforms and reporting dashboards supporting 1,000+ participant registrations per event.

## SIDE PROJECT

**Learnova — Multi-Tenant School Management Platform** · *NestJS 11* · *PostgreSQL/TypeORM* · *BullMQ* · *React (Turborepo)* · [github.com/adexy1/learnova](https://github.com/adexy1/learnova)

- Solo-building a school platform covering the full academic, administrative, and financial lifecycle: per-tenant database isolation, four switchable personas, automated report cards and broadsheets, fee invoicing with Paystack, and a complete payroll module (salary structures, runs, payslips, audit trail).
- NestJS 11 API (JWT refresh-token auth, Argon2, BullMQ workers, Paystack subscription billing); Turborepo frontend (TanStack Start SSR, Vite, Astro) with shared shadcn/ui, API-client, and type packages — RBAC + plan-entitlement gating, offline draft autosave, SSE notifications.
- Self-hosted in production on Oracle Cloud (Ampere ARM): Docker Compose orchestrating PostgreSQL, Redis, API, and SSR frontend behind one hardened Nginx entry point — wildcard TLS for tenant subdomains, only port 443 exposed, Cloudflare caching + WAF in front.

## EDUCATION & CERTIFICATIONS

**BSc Accounting** — University of Ilorin · 2021 — applied daily: payroll logic, tax computation, reconciliation, financial reporting

**MSc Information Technology** — Miva Open University · Expected 2026

HackerRank React Certification (2023) · Udemy — Testing React with Jest and React Testing Library (2024)